FitNourish AI – Intelligent Nutrition & Wellness Assistant

Project ID: 25-26J- 386

Project Proposal Report

Weerarathne H.P

B.Sc. (Hons) Degree in Information Technology Specialization in Information Technology

Department of Information Technology

Sri Lanka Institute of
Information Technology
Sri Lank

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Declaration

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I declare that this proposal is a result of our own work and has not been submitted for any degree or other programs at any other institution. I confirm, to the best of my knowledge, that it does not contain any material that is previously published or written material, except where due acknowledgement is made within the text of this report.

Name	Student ID	Signature
Weerarathne H. P	IT22907684	

As the supervisor of the above-mentioned students, I hereby confirm that they are undertaking their final year undergraduate research project under my supervision and guidance.

.... Supervisor's Signature

29/8/25 Date

i

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I would like to express my sincere gratitude to my supervisor, [Supervisor Name], for their invaluable guidance, support, and encouragement throughout the development of this project. Their expertise and constructive feedback were instrumental in shaping the direction and quality of this research.

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Finally, I extend my heartfelt appreciation to my family and friends for their continuous encouragement, patience, and motivation throughout this journey.

Abstract

The increasing prevalence of non-communicable diseases (NCDs) and multi-morbidity poses significant challenges to traditional "one-size-fits-all" dietary interventions, which often fail to account for individual physiology, lifestyle, and overlapping dietary restrictions. This research introduces **FitNourish AI**, an intelligent nutrition platform featuring the **Multi-Disease Adaptive Nutrition Engine (MDANE)** to address these limitations.

MDANE focuses on calculating safe and personalized daily nutrition targets by leveraging both **static user profile data** (age, gender, weight, height, BMI, and medical history) and **dynamic real-time biometric data** from smartwatches (steps, heart rate, activity level). It integrates a hybrid artificial intelligence framework that combines a **rule-based engine**, which encodes clinical guidelines to ensure safety for conditions such as diabetes and hypertension, with a **machine learning model (XGBoost)** that personalizes calorie and macronutrient requirements (protein, carbohydrates, fats). By merging profile and real-time data, the system creates a continuously adapting digital twin of the user, eliminating reliance on error-prone self-reported activity metrics.

Key innovations include a **multi-condition constraint matrix** for managing comorbid nutritional requirements, a **trigger-based recalculation mechanism** using cumulative sum (CUSUM) to detect metabolic shifts, and an **Explainable AI interface** that provides natural language justifications to enhance user trust and clinical transparency.

The study will validate the MDANE prototype through a controlled user study. Anticipated results include a significant improvement in **accuracy of nutrition target estimation** (MAE <10% for calorie prediction against ground-truthed expenditure data), high user satisfaction (SUS score >80), and strict adherence to clinical safety with **0% violation rate** of disease-related constraints. This represents a paradigm shift from generic diet recommendations to **dynamic, evidence-based, and safe personalized nutrition targets**. The platform has significant potential for applications in direct-to-consumer wellness services, clinical practice, and corporate health programs.

Keywords: *Personalized nutrition, nutrient target estimation, Machine learning, Explainable AI, Biometric monitoring*

Contents

List of Figures

List of Tables

List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
MDANE	Multi-Disease Adaptive Nutrition Engine
ML	Machine Learning
XAI	Explainable AI
SHAP	SHapley Additive exPlanations
CUSUM	Cumulative Sum (Control Chart)
BMR	Basal Metabolic Rate
TDEE	Total Daily Energy Expenditure
AEE	Actual Energy Expenditure (from wearables)
MET	Metabolic Equivalent of Task
PAL	Physical Activity Level
BMI	Body Mass Index
NCDs	Non-Communicable Diseases
WHO	World Health Organization
ADA	American Diabetes Association
AHA	American Heart Association
CKD	Chronic Kidney Disease

PCOS	Polycystic Ovary Syndrome
IBS	Irritable Bowel Syndrome
API	Application Programming Interface
JSON	JavaScript Object Notation
NLP	Natural Language Processing
NLG	Natural Language Generation
GUI	Graphical User Interface
B2C	Business-to-Consumer
B2B2C	Business-to-Business-to-Consumer
SaaS	Software-as-a-Service
KPI	Key Performance Indicator

List of Appendices

1. Introduction

The global burden of non-communicable diseases (NCDs) has reached epidemic proportions, with unhealthy diets established as a primary modifiable risk factor for mortality worldwide [1]. Conditions such as cardiovascular disease, type 2 diabetes, and hypertension collectively account for over 70% of global deaths, a crisis exacerbated by rising obesity rates and population aging [2]. Traditional nutritional interventions, including standardized dietary guidelines and static mobile applications, have proven inadequate to counter this trend. Their fundamental failure lies in a "one-size-fits-all" approach that ignores the complex interplay of individual genetics, microbiota, metabolism, and lifestyle [4], and—most critically—the pervasive challenge of multi-morbidity, where patients present with multiple, often conflicting chronic conditions [3].

FitNourish AI is conceived as a direct response to this critical gap in preventive healthcare. It is an intelligent, adaptive platform engineered to deliver **precision nutrition** through the seamless integration of artificial intelligence (AI) and real-time biometric monitoring. The cornerstone and primary innovation of this platform is the **Multi-Disease Adaptive Nutrition Engine (MDANE)**. MDANE is a sophisticated hybrid AI system responsible for the foundational task of calculating and continuously refining **dynamic nutritional targets**.

Unlike traditional calorie calculators [6], it synthesizes a comprehensive **user profile (age, gender, BMI, and medical history)** with continuous, high-frequency data streams from **wearable devices** [11], creating an evolving **digital twin** of the user's metabolic health [19]. By combining a **rule-based engine** (ensuring clinical safety for conditions such as diabetes and hypertension [10], [15]) with **machine learning models** (enabling personalization of calorie and macronutrient requirements [5], [16]), MDANE generates precise **calorie and macronutrient targets** that are both **aligned with user goals** and **safe under multiple medical comorbidities** [9], [14]. Key innovations include its ability to adapt to metabolic changes through concept drift detection [22] and to enhance trust and transparency via explainable AI [17], [23].

This represents a paradigm shift from static, generic diet recommendations to a **dynamic, evidence-based, and intelligent nutritional guidance system** that can support users across diverse healthcare contexts

1.1. Leveraging AI and Machine Learning for Personalized Nutrition Targets

The **Multi-Disease Adaptive Nutrition Engine (MDANE)** begins its hyper-personalization process by constructing a **robust digital phenotype** of the user. This phenotype is a comprehensive computational representation integrating multiple data domains—including demographic, anthropometric, physiological, lifestyle, and goal-oriented information—to enable **precise, safe, and individualized nutrition target estimation** [6], [11], [19]. Evidence shows that integrating both static characteristics (e.g., age, BMI, comorbidities) and dynamic behavioral and metabolic data improves accuracy in personalized nutrition outcomes [4], [16].

MDANE continuously learns from **longitudinal biometric data** collected via wearable devices. By performing **time-series analysis** of resting heart rate, heart rate variability, sleep patterns, and active energy expenditure, the engine detects subtle yet physiologically significant shifts in metabolism [11], [12].

For example, a sustained decrease in activity energy expenditure combined with a weight loss plateau may indicate metabolic adaptation, prompting MDANE to recalculate caloric targets. Such adaptive recalculations are supported by prior research demonstrating that dynamic monitoring provides more accurate estimations of individual energy requirements than static predictive equations alone [6], [24]. This creates a **predictive, closed-loop system**, enabling iterative optimization of calorie and macronutrient targets [19], [22].

1.1.1. User Profile Analysis: Constructing the Digital Phenotype

The digital phenotype consists of four primary data domains:

Domain	Examples	Purpose
Demographic & Anthropometric	Age, sex, height, weight, BMI	Establish baseline energy requirements using predictive equations such as Mifflin-St Jeor [6]
Physiological & Medical	Diagnosed conditions (T2DM, Hypertension, PCOS, IBS), medications, severity, duration	Populate the **multi-condition constraint matrix** in the rule- based engine
Lifestyle & Behavioral	Activity level, occupation, sleep patterns, dietary preferences/allergies	Provide context for estimating daily energy expenditure and adherence
Goal-Oriented	Target weight, pace of weight change, athletic performance goals	Guide caloric and macronutrient adjustments

1.The digital phenotype consists of four primary data domains

Data preprocessing:

- ❖ Continuous variables → **Z-score normalization**
- ❖ Categorical variables → **one-hot encoding**
- ❖ Missing values → **k-Nearest Neighbors (KNN) imputation** [7]

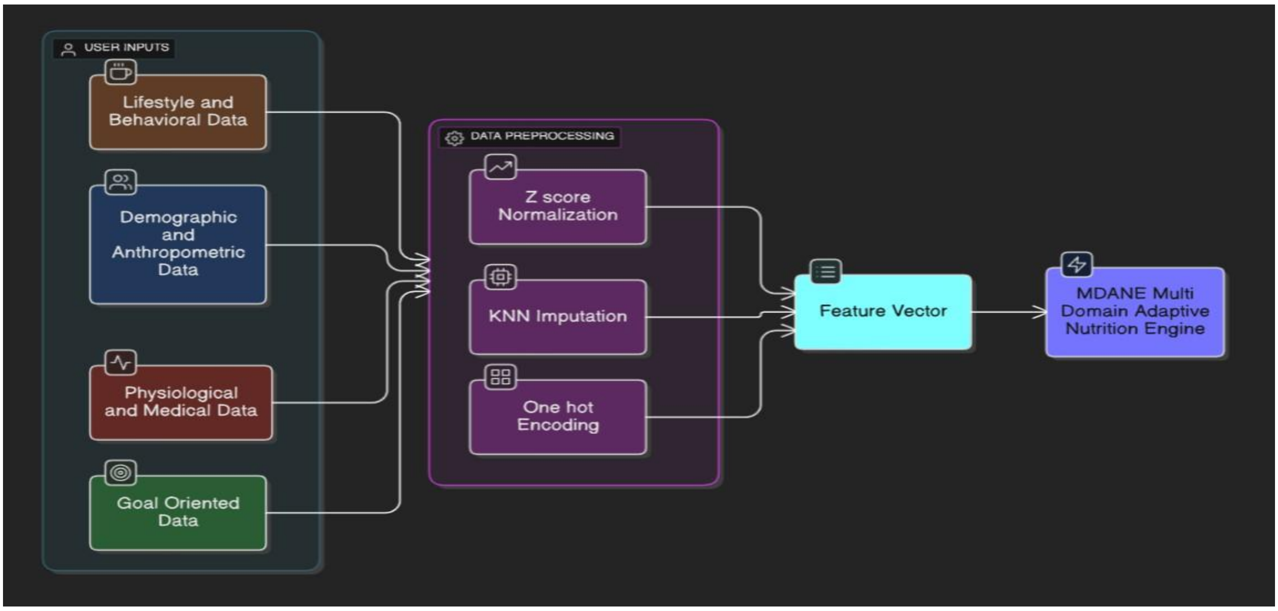

Figure 1.Data Preprocessing Workflow

1.1.2. Personalized Questionnaires

The accuracy of MDANE is strongly dependent on **input quality**. The onboarding questionnaire is carefully designed to maximize information gain while minimizing user burden.

Example: Questions about hypertension capture severity and medication use, informing sodium constraints.

Example: Fitness goals map directly to evidence-based macronutrient strategies (e.g., bodybuilding → high-protein target based on ISSN guidelines [8]).

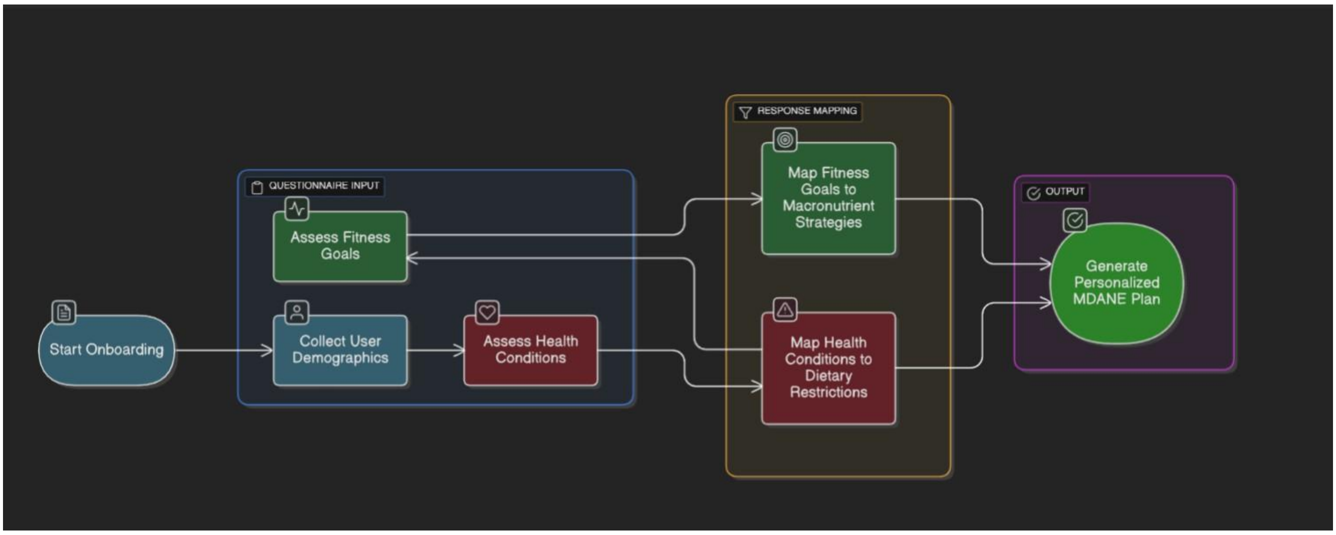

Figure 2.ersonalized Questionnaire Framework

1.1.3. Hybrid AI Architecture for Nutrition Target Optimization

MDANE uses a hybrid AI system, combining symbolic AI (rule-based) and sub-symbolic AI (machine learning), suitable for safety-critical healthcare applications [9].

Rule-Based Engine ("The Guardian")

- ❖ Enforces clinical guidelines from WHO [1], ADA [10], and AHA.
- ❖ Serves as a high-level constraint solver for multi-disease users: e.g., limiting sugar for diabetes, sodium for hypertension, or both for comorbid conditions.
- ❖ Ensures safety: any ML-generated target violating rules is overridden [15].

Machine Learning Layer ("The Optimizer")

- ❖ Operates within rule-based boundaries to optimize targets.
- ❖ Uses supervised regressors (e.g., XGBoost) [16] to personalize:
- ❖ Caloric deficits for weight loss (e.g., 500–750 kcal)
- ❖ Macronutrient splits based on satiety, energy, and metabolic response
- ❖ Trains on outcomes such as successful weight loss or improved glycemic control [16].

1.1.4. Dynamic Adaptation and Feedback

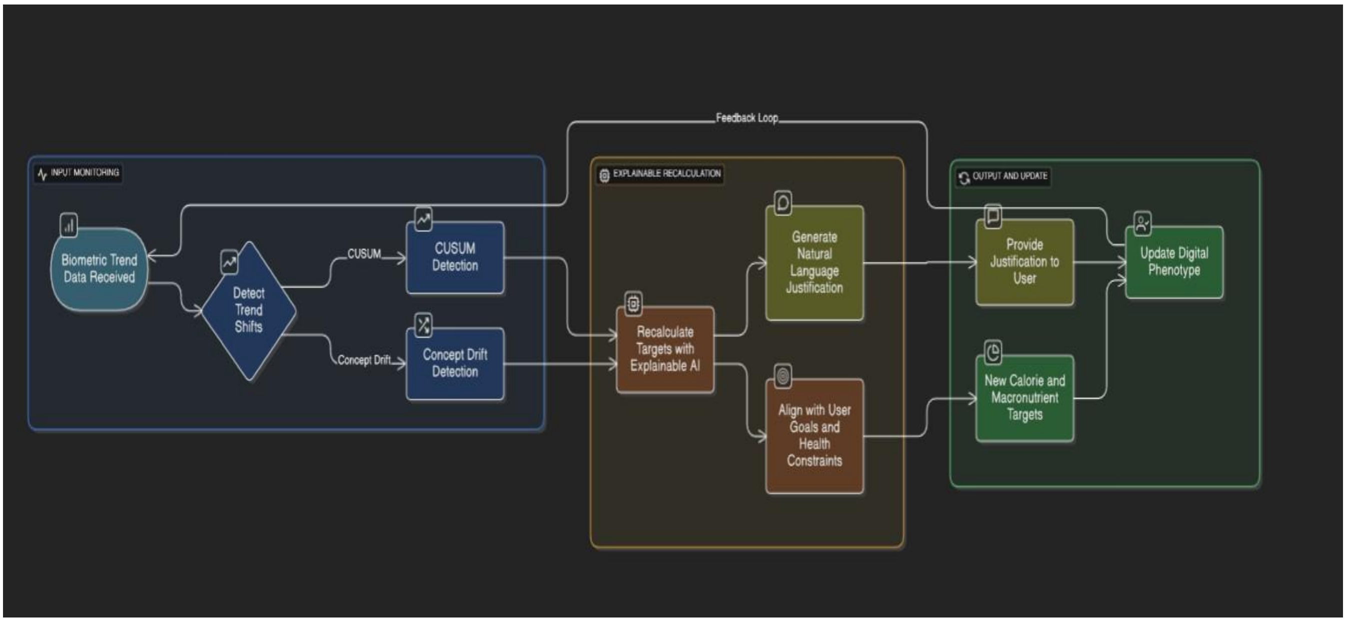

Figure 3.Adaptive Feedback Framework in MDANE

- **Continuous recalculation:** MDANE monitors trends via **CUSUM and concept drift detection** [22].
- **Explainable AI:** Outputs provide **natural language justifications**, increasing user trust and clinical transparency [17], [23].
- **Outcome:** Predictive, continuously improving calories and macronutrient targets aligned with user goals and health constraints.

2. Background

2.1 Traditional Approaches to Diet Planning

Historically, nutritional science and dietetics have relied on reductionist methodologies rooted in population-level epidemiology. The clinical gold standard for calculating energy needs has involved the use of predictive equations such as Harris-Benedict or, more recently and accurately, Mifflin-St Jeor, to estimate Basal Metabolic Rate (BMR)—the energy expended at complete rest [6]. This BMR is then multiplied by a Physical Activity Level (PAL) multiplier, a generic factor (e.g., 1.2 for sedentary, 1.55 for moderately active) to estimate Total Daily Energy Expenditure (TDEE) [11].

While these equations are validated at a population level, they exhibit significant error at the individual level, often underestimating or overestimating true energy needs by 20% or more [12]. This error is compounded by the well-documented inaccuracy of self-reported activity levels, which are plagued by recall bias and social desirability bias [13].

Macronutrient distribution in these traditional systems often follows rigid, percentage-based paradigms (e.g., the 40-30-30 "Zone" diet). These approaches are physiologically flawed because the body requires absolute amounts of nutrients (grams of protein per kg of body weight) rather than ratios relative to total caloric intake [8]. Furthermore, they are utterly inadequate for managing multi-morbidity, where macronutrient priorities often conflict. For example, a low-carbohydrate diet beneficial for glycemic control in diabetes may inadvertently increase saturated fat intake, potentially exacerbating dyslipidemia [14].

The static nature of these plans is their ultimate failure; they are unable to adapt to the user's changing weight, body composition, fitness level, or metabolic health, rendering them ineffective and potentially demotivating over the long term.

Justification for MDANE:

These shortcomings demonstrate the necessity for an adaptive system like **MDANE**, which goes beyond population averages by leveraging **real-time BMI, body composition, and wearable data**. Unlike static models, MDANE dynamically recalibrates to individual physiology and evolving health conditions, ensuring continuous accuracy and relevance.

2.2 The Paradigm Shift: Advances in AI and Machine Learning for Nutrition

The application of artificial intelligence in nutrition represents a seismic shift from generic to personalised and from static to adaptive systems. Early research focused on expert systems that used simple "if-then" rule sets, but these lacked the ability to learn from data [15]. The current wave of innovation is fueled by machine learning:

- **Predictive Modelling:** Landmark studies, such as the one by Zeevi et al. [16], used ML to predict individual postprandial glycemic responses to meals based on personal features like gut microbiome composition, demonstrating that personalized diets achieved significantly better outcomes than standardized ones.

- **Natural Language Processing (NLP):** Transformer-based models like BERT are being fine-tuned on scientific literature to keep dietary recommendation engines updated with the latest evidence, a field known as literature-based discovery [17].

- **Computer Vision:** Deep learning models like Convolutional Neural Networks (CNNs) are achieving high accuracy in automated food recognition and portion size estimation from images, tackling the long-standing problem of inaccurate dietary logging [18].

However, a significant challenge persists: the trade-off between personalisation and safety. Purely data-driven ML models can make recommendations that are statistically optimal but clinically dangerous—for example, suggesting intermittent fasting for a diabetic on certain medications.

This has led to a growing consensus in biomedical informatics around the necessity for hybrid AI systems that can integrate data-driven learning with knowledge-driven reasoning, a gap that this research directly addresses [9].

Justification for MDANE:

This gap highlights the need for **hybrid AI systems** that combine **data-driven personalization** with **knowledge-based medical reasoning**. MDANE embodies this approach by integrating ML predictions with established clinical rules, ensuring that personalization remains safe, explainable, and effective across diverse health conditions.

2.3 The Complex Challenge of Multi-Disease Nutritional Management

Nutritional management for individuals with comorbidities is a complex problem of multi-objective optimization, where clinical guidelines for single diseases can provide directly conflicting advice. For instance:

- **Diabetes + Chronic Kidney Disease (CKD):** Diabetes management encourages adequate protein intake for satiety and glycemic control, while CKD management often requires protein restriction to slow the progression of nephropathy.

- **Hypertension + Osteoporosis:** A DASH diet for hypertension is rich in fruits and vegetables, which can be beneficial for bone health. However, if it is also high in potassium and phosphorus, it may be contraindicated for patients with advanced CKD.

- **Obesity + Heart Disease:** Aggressive caloric restriction for obesity must be carefully managed to ensure adequate intake of essential nutrients to support cardiovascular health.

Existing digital solutions typically handle this complexity poorly, either by focusing on a single "primary" condition or by applying the most restrictive rules from all conditions, which can lead to an overly restrictive, unpalatable, and potentially nutrient-deficient diet.

There is a pronounced lack of intelligent systems that can perform constraint-based reasoning and Pareto optimization to find a feasible and optimal nutritional plan that satisfies all conditions simultaneously, making this a critically important and underexplored area for research.

Justification for MDANE:

MDANE addresses this challenge through **constraint-based reasoning and Pareto optimization**, enabling it to balance conflicting dietary needs across multiple diseases. Instead of oversimplifying or over-restricting, MDANE generates **clinically viable, balanced nutrition plans** tailored to the individual’s complete health profile.

2.4 Digital Twin Integration: A Novel Paradigm for Dynamic Health Modeling

The concept of a "digital twin"—a dynamic, virtual representation of a physical system that is continuously updated with real-time data—is revolutionary for personalized health [19]. In the context of FitNourish AI, the MDANE constructs a functional digital twin of the user's metabolism. The initial twin is built from the static user profile. This twin is then continuously updated with real-time data streams from wearables: heart rate and heart rate variability provide insights into autonomic nervous system function and recovery status; sleep stage data offers a proxy for hormonal regulation (cortisol, growth hormone); and activity calories provide a ground-truth measure of energy expenditure.

This living digital model allows for simulation and "what-if" analysis. The AI can, in principle, simulate the impact of a specific dietary intervention on the twin—predicting its effect on weight, blood glucose trends (if data is available), and other derived biomarkers—before recommending it to the real user. This moves the platform from being reactive (responding to logged data) to being predictive and preventive, a cornerstone of future healthcare.

While full predictive simulation may be a longer-term goal, the foundational architecture of MDANE, particularly its storage of time-series data and its recalibration mechanism, is designed to support this evolution.

Justification for MDANE:

By embedding digital twin integration, MDANE becomes a **proactive nutrition system**—not only tracking current health but also simulating outcomes of dietary choices. This predictive capability ensures that interventions are safe, personalized, and future-proof, positioning MDANE as a transformative solution in adaptive nutrition.

2.5. Summary

Traditional diet planning is increasingly inadequate for modern health challenges, particularly the rise of multi-morbidity. While AI and machine learning offer promising solutions, their adoption has been limited by safety and explainability concerns. The integration of hybrid AI architectures with the digital twin concept presents a novel path forward. MDANE is not merely an incremental improvement; it represents a synthesis of these advanced technologies, creating a truly adaptive, safe, and effective personalised nutrition system.

This table contrasts the core philosophies, highlighting MDANE's transformative potential.

Feature	Traditional Approach	MDANE (Adaptive & Hybrid AI) Approach
Core Philosophy	Reductionist, population-based	Holistic, individual-centric
Energy Calculation	Static equations (e.g., Mifflin-St Jeor) with generic multipliers	Dynamic calibration using real-time data (wearables, progress)
Macronutrient Strategy	Rigid percentages (e.g., 40-30-30)	Absolute, condition-aware gram targets (e.g., protein g/kg)
Disease Management	Single condition or most restrictive rule	Multi-condition constraint optimization (Pareto optimality)
Adaptability	Static plan, unchanged until review	Continuous, dynamic adaptation (Digital Twin)
Basis for Decisions	Population averages & static guidelines	Personal data + Clinical knowledge (Hybrid AI)
Key Limitation	High individual error, unsafe for multi-morbidity	Addresses these limitations directly

2.Paradigm Shift from Traditional to MDANE Approac

This table illustrates the complex optimization challenge MDANE must solve, justifying its rule-based constraint engine.

Condition 1	Condition 2	Nutritional Conflict	MDANE's Resolution Strategy
Diabetes (T2DM)	Chronic Kidney Disease (CKD)	T2DM: Needs adequate protein for satiety/glycemia. CKD: Requires protein restriction.	Optimizes for minimal effective protein to meet T2DM goals without exceeding CKD limits.
Hypertension	Osteoporosis	Hypertension: DASH diet (high in K+, Mg2+). Advanced CKD: Requires potassium (K+) restriction.	Prioritizes low-sodium, bone-healthy nutrients while strictly bounding potassium based on CKD severity.
Heart Disease	Obesity	Obesity: Requires caloric deficit. Heart Disease: Needs nutrient-dense, heart-healthy foods.	Ensures aggressive caloric targets still meet minimum thresholds for essential fats, fiber, and micronutrients.

3.Paradigm Shift from Traditional to MDANE Approach

This diagram visualizes the core innovation of MDANE: moving from a static profile to a dynamic, predictive digital twin.

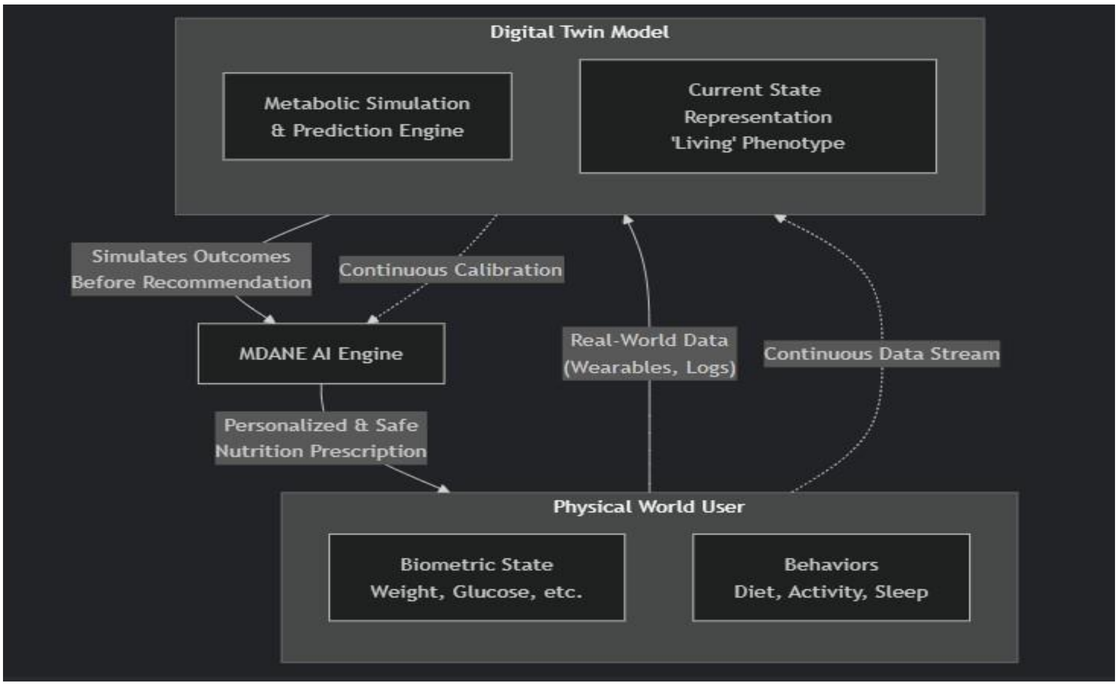

Figure 4.The Digital Twin Feedback Loop for Dynamic Adaptation

3. Literature Review

3.1. Foundational Challenges in Personalized Nutrition Target Generation

The automated generation of personalised nutrition targets is a complex interdisciplinary challenge spanning computer science, nutrition, and clinical medicine.

❖ Data Heterogeneity and Integration:

- Inputs span multiple temporal scales: static user profile, episodic meals, and high-frequency wearable sensor data [20].
- Requires robust pipelines for temporal alignment, noise reduction, and feature extraction. Techniques include **dynamic time warping** and **rolling-window aggregation**.

❖ Multi-Objective Optimization:

- Conflicting objectives: maximize satiety vs minimize calories, optimize muscle synthesis vs manage blood glucose, nutrient density vs budget [21].
- Requires **Pareto-optimal solutions** using algorithms like **NSGA-II** (Non- dominated Sorting Genetic Algorithm II).

❖ Dynamic Adaptation and Concept Drift:

- Human metabolism adapts over time (adaptive thermogenesis), health fluctuates [22].
- Models must detect drift via prediction error monitoring or data distribution shifts and adjust dynamically.

❖ Explainability and Trust:

- Health-critical systems cannot rely on black-box models.
- Use of **XAI techniques** such as **LIME** and **SHAP**, with integration into **natural language explanations** for end users [23].

Justification for MDANE:

These foundational challenges underscore the necessity of a system like **MDANE**, which is explicitly designed to:

- **Integrate heterogeneous data** from user profiles, BMI, and wearable devices,
- Perform **multi-objective optimization** across comorbidities and dietary goals,
- **Adapt dynamically** to metabolic changes and concept drift, and
- Provide **explainable, clinically safe nutrition targets** for multi-disease users.

Without such an engine, conventional or purely ML-based approaches cannot deliver personalized, safe, and actionable recommendations. MDANE's hybrid AI design ensures both **precision and clinical accountability**, making it critical advancement in personalized nutrition target generation.

This diagram visualizes the core innovation of MDANE: moving from a static profile to a dynamic, predictive digital twin.

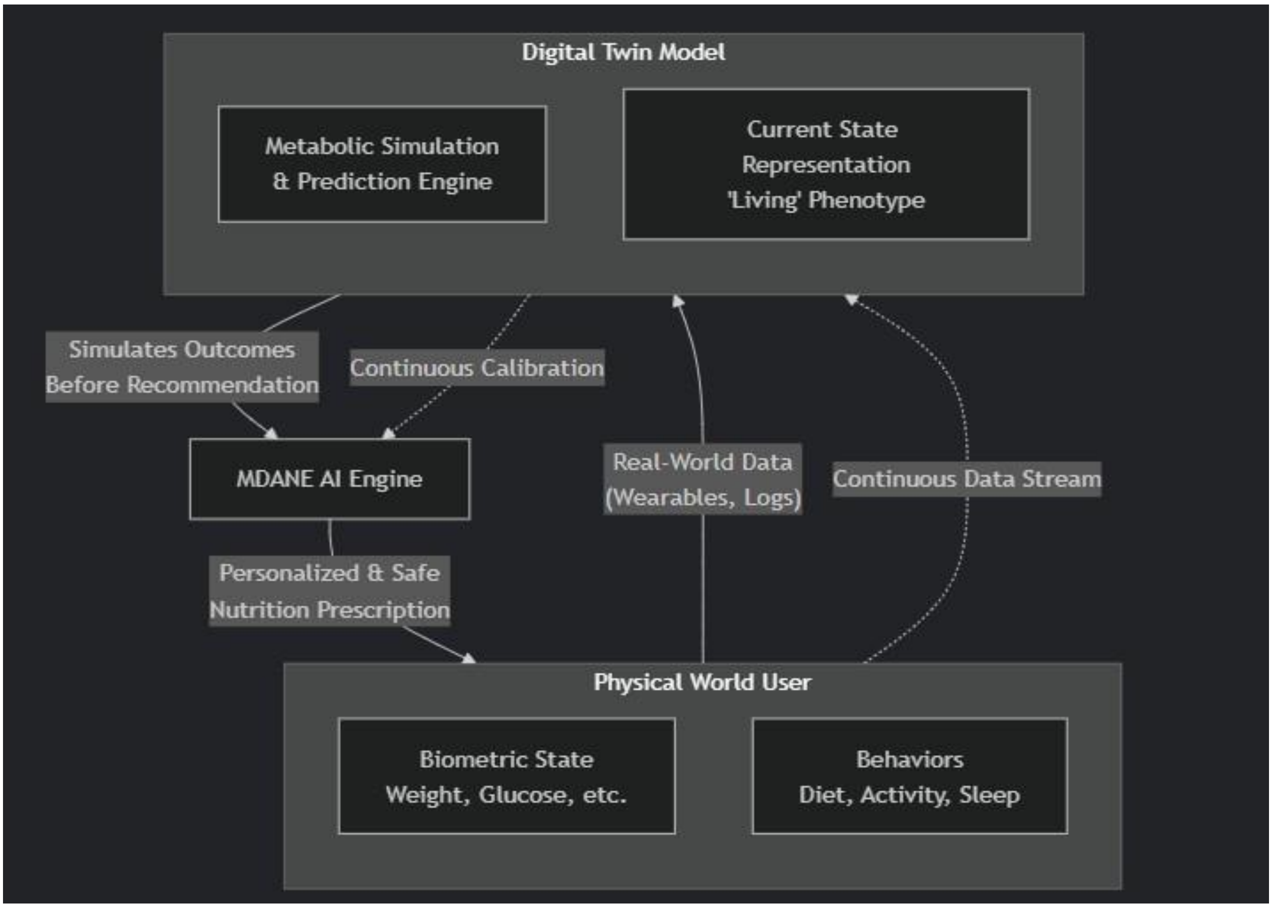

Figure 5.The Digital Twin Feedback Loop for Dynamic Adaptation

3.2. Critical Analysis of Existing Work

3.2.1. Calculating Energy Requirements: The Limitations of BMI and PAL

- **BMI issues:** Does not differentiate fat vs lean mass; misclassifies “skinny fat” individuals [24].
- **PAL multipliers:** Coarse, generic; fail to capture inter-individual variability.
- **Evidence:** Doubly labelled water studies show large differences in energy expenditure for same activities [12].
- **MDANE relevance:** Uses **wearable-derived, objective data** to calibrate personalized energy needs.

Table: Traditional vs. MDANE Approach to Energy Requirements

Feature	Traditional Model (BMI + PAL)	MDANE's Adaptive Model
Body Composition	**Relies on BMI:** • Misclassifies muscular individuals as overweight • Misclassifies "skinny fat" (normal weight obese) as healthy	**Seeks Body Composition Data:** • Prioritizes body fat % from scans or smart scales • Uses fat-free mass for more accurate calculations
Activity Assessment	**Uses Generic PAL Multipliers (e.g., 1.2-1.9):** • Based on self-reported activity (highly inaccurate) • Fails to capture daily variability and NEAT	**Uses Objective Wearable Data:** • Calibrates using real-time calorie burn from devices • Continuously adjusts based on actual daily movement
Scientific Basis	Population-level averages with high individual error (±20-30%)	**Personalized, data-driven calibration** grounded in individual reality
Key Weakness	**Static and Coarse:** A "set-and-forget" calculation that becomes outdated.	**Dynamic and Precise:** A "living" estimate that evolves with the user.
Evidence Against	Doubly labelled water studies show vast energy expenditure variation for the same PAL category [12].	Aligns with the gold-standard (doubly labelled water) by using direct, individual measurement.

4.Traditional vs. MDANE Approach

3.2.2. Macronutrient Allocation: From Fixed Ratios to Absolute Goals

- **Observation:** Traditional diet apps use fixed macronutrient ratios (e.g., 40-30-30), which ignore individual physiological needs.
- **Literature Finding:** Contemporary nutritional science emphasizes absolute nutrient targets (grams per kg body weight), especially for protein [8], [25].
- Algorithmic/**MDANE Approach:** MDANE calculates goal-oriented absolute targets and integrates quality constraints (e.g., glycemic index, fibre) through the rule engine, improving alignment with evidence-based nutrition [26].

MDANE is an application of this validated and responsible paradigm to the domain of multi-disease nutritional planning. Multi-Disease Adaptive Nutrition Engines

3.2.3. The Case for Hybrid ML + Rule-Based Systems in Healthcare

- **Observation:** Purely ML systems can be unsafe in health-critical applications due to lack of clinical knowledge integration.
- **Literature Finding:** Studies show hybrid AI (ML + rule-based) ensures safety and adaptability. Examples:
    1. Wiens et al. [9]: Responsible ML framework for healthcare.
    2. Nemati et al. [27]: Hybrid RL system for ICU dosing with safety-critical rule constraints.
- **Algorithmic/MDANE Approach:** MDANE uses a hybrid architecture where the rule-based engine ensures safety for multi-disease constraints, and the ML model personalizes nutritional targets

3.2.4. The Vacuum in Multi-Disease Adaptive Nutrition Engines

- **Observation:** Existing commercial systems focus on single goals or single diseases; multi-disease interactions are poorly handled.
- Literature **Finding:** No validated research prototypes address combinatorial dietary constraints across multiple conditions.
- Algorithmic/**MDANE Approach:** MDANE introduces a **multi-condition constraint matrix**, inspired by constraint satisfaction problems (CSP) [28], to codify interactions between overlapping nutritional requirements.

3.3. Emerging Trends and Future Directions

1. Explainable **AI (XAI)**
   - **Challenge:** Users cannot interpret Blackbox ML outputs.
   - **Literature Finding:** SHAP values are useful but not actionable for end-users [23].
   - **MDANE Approach:** Natural language explanations translate model insights into understandable and actionable guidance.
2. Federated **Learning (FL)**
   - **Challenge:** Privacy concerns when aggregating user data for ML training.
   - **Literature Finding:** FL allows model training without sharing raw data [29].
   - **MDANE Approach:** Enables continuous learning from multiple users while maintaining privacy.
3. Integrative **Omics: Nutrigenomics and Microbiome**
   - **Challenge:** Individual genetic and microbiome data affect nutrient metabolism.
   - **Literature Finding:** Nutrigenomics and microbiome studies reveal personalized nutrient response [30].
   - **MDANE Approach:** Digital twin architecture is designed to integrate future omics data for enhanced personalization.

4. Research Gap

Based on the critical analysis of existing literature and commercial solutions, a significant and unaddressed research gap is identified. There is a lack of an integrated, intelligent system that **simultaneously**:

1. **Dynamically adapts** nutritional targets using **real-time, objective biometric data** from wearables to move beyond static predictive equations and self-reported data.

2. Incorporates a **robust, rule-based safety layer** to manage **multi-morbidity** through a formal constraint-solving approach, ensuring clinical safety for users with complex, co-existing health profiles.

3. Provides **transparent, explainable recommendations** through user-centric natural language justifications to build trust and facilitate collaboration with healthcare providers.

Existing solutions address, at best, one or two of these aspects in isolation but fail to integrate all three into a cohesive, scalable, and clinically robust engine. This project aims to bridge this critical gap through the design, development, and validation of the Multi-Disease Adaptive Nutrition Engine (MDANE).

5. Research Problem

The core research problem this project seeks to address is therefore formulated as follows:

"How can a hybrid artificial intelligence system integrate a rule-based engine for clinical safety with a machine learning model for personalization to generate dynamic, personalized nutritional targets? How can this system continuously adapt to a user's evolving physiological state from smartwatch data while respecting the complex and often conflicting constraints of multiple chronic health conditions?"

6. Objectives

6.1. Main Research Objective

To design, develop, and empirically validate a Multi-Disease Adaptive Nutrition Engine (MDANE) that generates safe, effective, and explainable dynamic nutritional recommendations by leveraging a hybrid AI architecture operating on comprehensive user profiles and real-time biometric data.

6.2. Specific Research Objectives

To design and implement a scalable, robust data pipeline for the acquisition, cleaning, validation, and fusion of heterogeneous data from user profiles (structured) and smartwatch APIs (Google Fit, Apple HealthKit - time-series).
To design, codify, and implement a multi-condition rule-based engine based on clinical guidelines from the WHO, ADA, and AHA, formalised as a configurable constraint matrix, to ensure the absolute safety of all generated recommendations for users with comorbidities.

To train, validate, and deploy a supervised machine learning model (XGBoost Regressor) to predict personalised caloric and macronutrient targets within the constraints defined by the rule engine, using both initial user data and ongoing biometric trends.

To develop and integrate a dynamic recalculation module that uses both time-based (bi-weekly) and event-based triggers (using CUSUM control charts for change point detection) on aggregated user data to automatically initiate adjustments to the nutrition plan.

To implement an Explainable AI (XAI) interface using SHAP values and natural language generation techniques to produce clear, auditable, and user-friendly justifications for every recommendation provided by the system.

7. Methodology

This chapter details the architectural design, core components, and underlying algorithms of the Multi-Disease Adaptive Nutrition Engine (MDANE). It provides a technical deep dive into the data flow, processing logic, and the integration of rule-based and machine learning systems that enable personalized, adaptive, and explainable nutritional guidance.

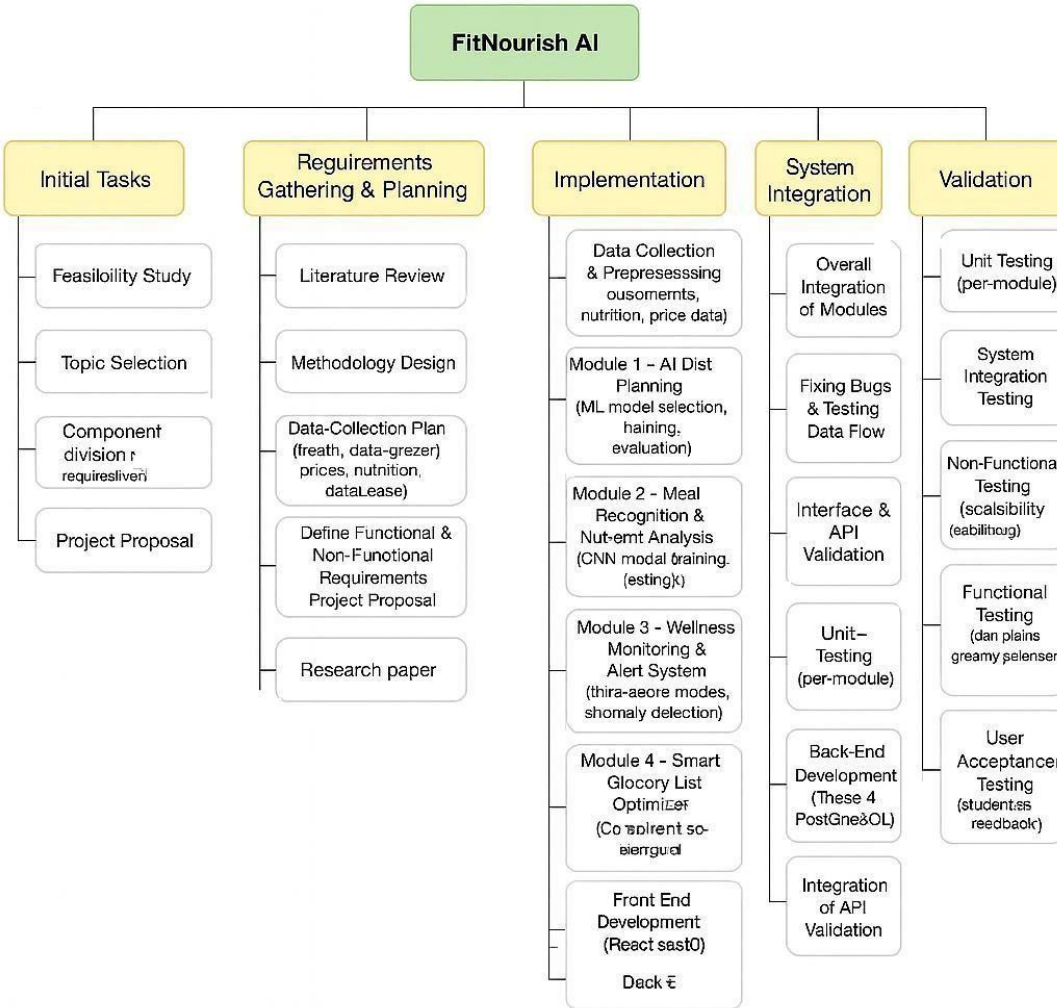

Figure 6.FitNourish AI Development Lifecycle

7.1. SysteSystem Overview and High-Level Architecture

The MDANE system is designed as a modular, scalable, and secure microservices architecture. The data flow follows a structured pipeline pattern to ensure robustness, clarity in processing stages, and ease of maintenance. The high-level architecture is depicted in Figure 7.1 and described in detail below.

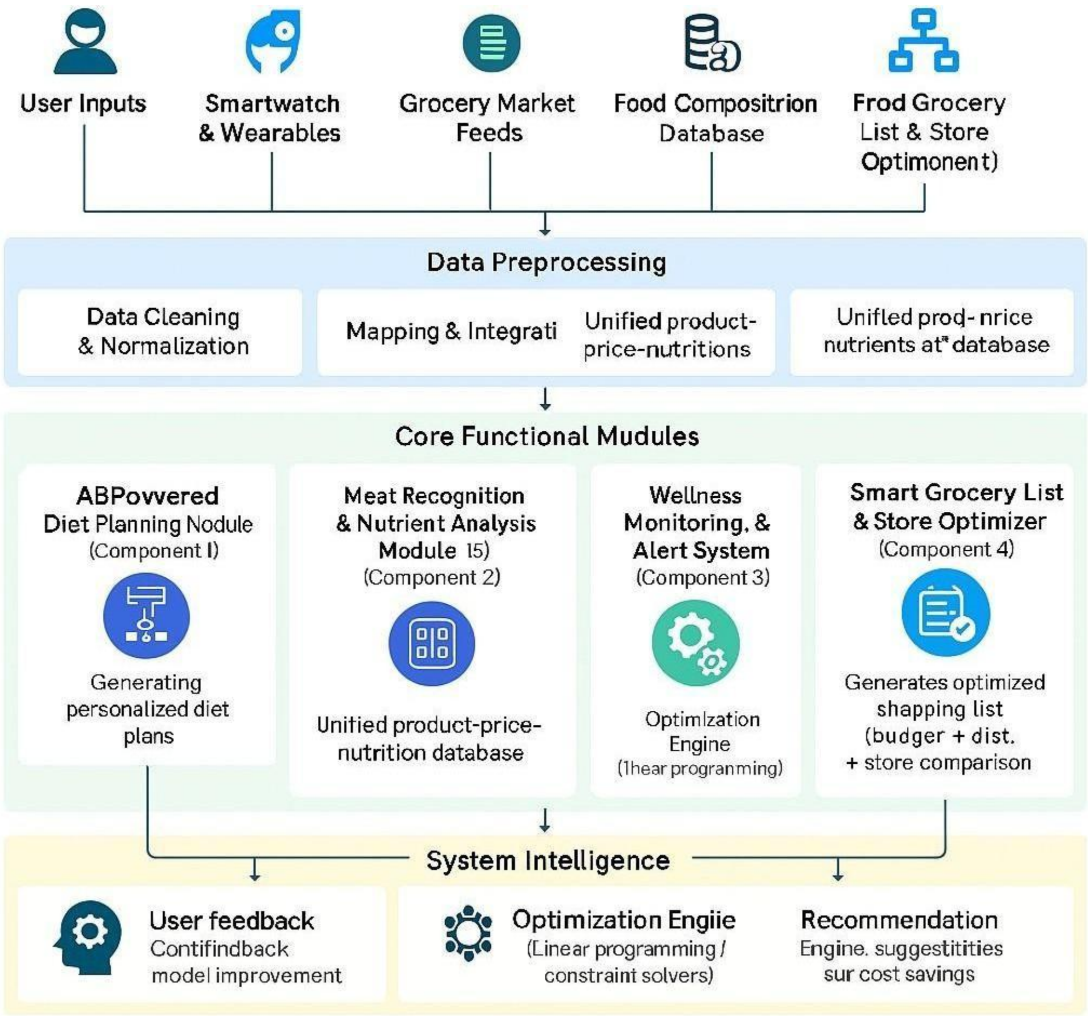

Figure 7.High-Level System Architecture Diagram

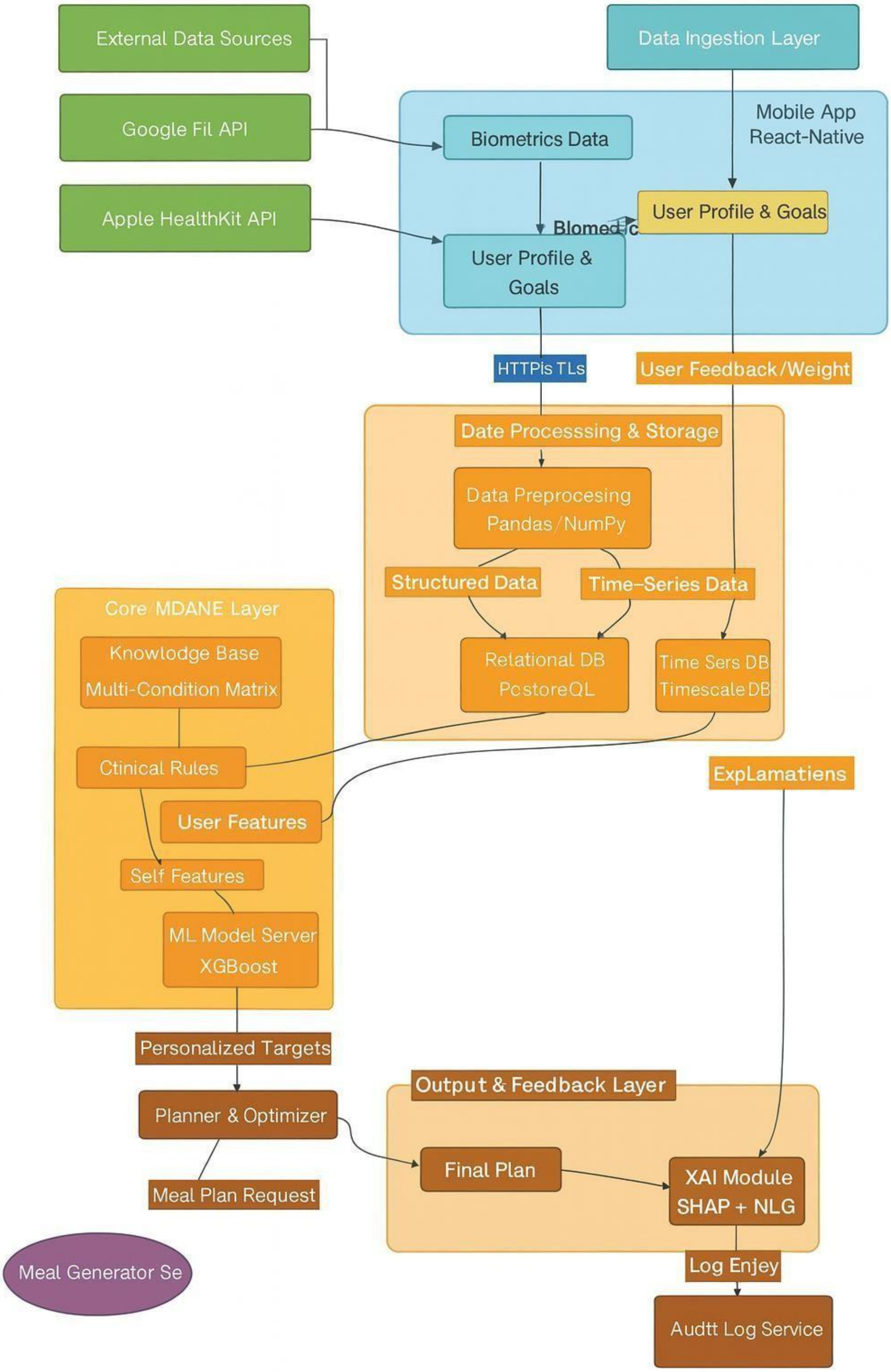

Figure 8.MDANE Data Flow and Processing Pipeline

Data Ingestion Layer: This is the user-facing entry point of the system.

- **Mobile Application (GUI):** Provides the user interface for manual data entry (e.g., age, weight, conditions, goals) and displays the generated plans and explanations.

- **API Connectors:** Secure backend services that handle OAuth 2.0 authentication and data retrieval from **Google Fit** and **Apple HealthKit** APIs. All data transmission is encrypted via HTTPS.

Data Processing & Storage Layer: This layer is responsible for transforming and persisting user data.

- **Preprocessing Pipelines:** Python scripts utilizing **Pandas** and **NumPy** for data cleaning, normalization, handling missing values, and feature engineering (e.g., calculating 7-day rolling averages for biometrics).

- **Primary Database:** A **PostgreSQL** relational database stores structured, non-time-series user data (user profiles, conditions, goals, generated plans, audit logs).

- **Time-Series Database:** A **TimescaleDB** (a PostgreSQL extension) hypertable is optimized for efficiently storing and querying high-volume, timestamped biometric data (heart rate, step count, active energy).

Core MDANE Layer: The heart of the system, a hybrid AI engine.

- **Knowledge Base (Multi-Condition Matrix):** A dedicated table in PostgreSQL storing all nutritional constraints and rules.

- **Rule-Based Engine:** The first stage of processing. It takes user data, applies all relevant constraints from the Knowledge Base, and outputs a "clinically safe" feature set.

- **ML Model (XGBoost):** The second stage. It takes the rule-compliant feature set and personalizes the caloric and macronutrient targets based on learned patterns from similar users.

Output Layer: Formats and delivers the final result.

- **The Planner:** An optimization module that translates the final targets into a structured meal plan, ensuring it meets all rules and is portioned correctly across meals.

- **XAI (Explainable AI) Module:** Generates natural language explanations for the recommendations using SHAP values and feature importance.

- **JSON API Response:** The Planner and XAI output are formatted into a standardized JSON response for the Mobile Application and the Meal Generator service.

Feedback Loop: Enables the system's adaptive capability.

- User interactions (logging meal compliance, updating weight) and new biometric data from smartwatches are fed back into the Data Processing Layer.

- This new data triggers the **Dynamic Recalculation** module, allowing the model to learn and adapt its future recommendations to the user's observed behavior and physiological changes.

7.2. Component Overview

The MDANE system comprises several interconnected software components:

- **User Management Service:** Handles authentication, profile creation, and updates.
- **Biometric Data Ingestion Service:** Orchestrates the periodic fetching of data from health APIs.
- **Calculation Engine (Core MDANE):** A dedicated service containing the Rule Engine, ML Model, and Planner logic.
- **Meal Generation Service:** An external service (consumed via API) that translates nutrient targets into specific food items and recipes.
- **Audit & Logging Service:** Records all system actions, decisions, and user feedback for traceability and model improvement.

7.3. Multi-Disease Adaptive Nutrition Engine (MDANE)

7.3.1. User Data Acquisition & Preprocessing4

Data is acquired from two primary sources:

1. **Manual Input:** Collected via the mobile app GUI upon user registration and during profile updates. This includes static and semi-static data: age, sex, height, weight, goal (e.g., weight loss, muscle gain), and medical conditions (e.g., Diabetes, Hypertension).

2. **Automated Biometric Input:** Continuously synced from Google Fit/Apple HealthKit. This includes dynamic, time-series data: step_count, heart_rate, active_energy_burned, sleep_duration.

Preprocessing Pipeline:

- **Cleaning:** Removal of physiologically impossible outliers (e.g., heart rate > 220, step count > 50,000 in an hour).

- **Imputation:** For missing numeric values (e.g., a gap in heart rate data), linear interpolation is used for short gaps. For longer gaps, a 7-day rolling average is applied.

- **Feature Engineering:** Creation of derived features that are more informative for the model:

  - resting_heart_rate_7d_avg: The average resting heart rate over the past week.
  - sleep efficiency: (time asleep / time in bed) * 100.
  - activity_calories_7d_avg: The average daily active energy expenditure over the past week.
  - weight_7d_trend: The slope of a linear regression fitted to the last 7 days of weight data.

7.3.2. Smartwatch & Biometric Integration

The system employs dedicated backend workers (e.g., **Celery** tasks) that run periodically to:

- ❖ Use OAuth 2.0 refresh tokens to securely access user data from the health APIs.
- ❖ Fetch data for the required time range (e.g., the last 24 hours).
- ❖ Transform the vendor-specific JSON responses into a standardized internal schema.
- ❖ Write this structured data to the Timescale DB hyper table for efficient time-series querying.

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- ❖ Transform the vendor-specific JSON responses into a standardized internal schema.
- ❖ Write this structured data to the Timescale DB hyper table for efficient time-series querying.

7.3.3. BMI Calculation and Categorization

The Body Mass Index (BMI) is calculated as a foundational metric for initial goal setting and risk assessment.

- ❖ **Formula:** $\text{BMI} = \text{weight (kg)} / (\text{height (m)})^2$
- ❖ **Categorization:** The calculated BMI is categorized according to WHO standards:
  - Underweight: $\text{BMI} < 18.5$
  - Normal weight: $18.5 \leq \text{BMI} < 25$
  - Overweight: $25 \leq \text{BMI} < 30$
  - Obesity Class I: $30 \leq \text{BMI} < 35$
  - Obesity Class II: $35 \leq \text{BMI} < 40$
  - Obesity Class III: $\text{BMI} \geq 40$

This category, combined with the user's stated goal, informs the initial caloric deficit/surplus strategy in the subsequent computation stage.

7.3.4. Calorie and Macronutrient Target Computation

The computation is a calibrated, multi-stage pipeline:

- **Baseline Calculation:** BMR is calculated using the Mifflin-St Jeor equation [6]. TDEE is estimated using MET (Metabolic Equivalent of Task) values from the compendium by Ainsworth et al. [11] to assign an activity multiplier, providing a more nuanced estimate than generic PAL values.

- **Biometric Recalibration (Closed-Loop Feedback):** This is a key innovation. The system calculates the user's Actual Energy Expenditure (AEE) from the smartwatch's active_energy_burned metric, averaged over a 7-day rolling window.

A calibration factor C is computed:

C = AEE_7day_avg / (TDEE_estimated - BMR). This factor is then used to calibrate the initial TDEE estimate: TDEE_calibrated = BMR + (C * (TDEE_estimated - BMR)). This TDEE_calibrated becomes the new baseline for the goal-based adjustment, ensuring the plan is continuously aligned with the user's real- world activity.

- **Macronutrient Allocation:** Uses absolute gram goals:
  - **Protein:** Set based on goal and body weight: 1.6-2.2 g/kg for muscle gain, 1.6-2.0 g/kg for weight loss [25].
  - **Fats:** Set to a minimum of 0.8-1.0 g/kg for essential fatty acids and hormonal health.
  - **Carbohydrates:** Calculated as: Carbs_g = (TDEE_calibrated - (Protein_g * 4) - (Fat_g * 9)) / 4. The rule engine then applies quality constraints (e.g., >30g fiber, low GI sources).

7.3.5. Multi-Condition Matrix Logic & Rule Engine

The matrix is implemented as a database table with a schema designed for flexible rule definition. The rule engine uses a forward-chaining inference system. It checks all conditions present in the user's profile and compiles a unified set of constraints.

- **Conflict Resolution:** If conflicts arise (e.g., two conditions demand different min/max values for the same nutrient), the engine employs a priority system (defined in the matrix) and defaults to the most conservative (safest) value. This conflict and its resolution are explicitly flagged for review in the audit log.

- **Example Matrix Schema and Entry:**

Conditions	Priority	Nutrient	Constraint_Type	Value	Unit	Rationale & Source
Diabetes	High	Added Sugar	Max	25	g/day	ADA 2023 Guidelines: Limit to reduce risk of CVD.
Hypertension	High	Sodium	Max	1500	mg/day	AHA Recommendation: For significant blood pressure lowering.
Diabetes, Hypertension	Critical	Added Sugar	Max	25	g/day	ADA & AHA: Combined constraint takes strictest value.
CKD (Stage 3-4)	High	Protein	Max	0.8	g/kg/day	NKF KDOQI Guidelines: To slow progression of kidney disease.
CKD (Stage 3-4)	Critical	Potassium	Max	3000	mg/day	NKF KDOQI Guidelines: To prevent hyperkalemia.
Goal: Weight Loss	Medium	Dietary Fiber	Min	30	g/day	ACSM/AND Guidelines: To promote satiety and gut health.
Irritable Bowel Syndrome (IBS)	Medium	FODMAPs	Avoid	-	-	Monash University Guidelines: Avoid high-FODMAP foods during flare-ups.

5.Matrix Schema

7.3.6. Machine Learning Personalization Layer

An **XGBoost Regressor** is chosen for its performance, speed, and built-in support for handling missing values and feature importance.

- **Training Data:** The model will be pre-trained initially on a synthetic dataset generated based on public data from NHANES [31], before being fine-tuned with real, anonymized user data (with ethical approval).

- **Features (X):** Encoded user profile (age, sex, goal, conditions), aggregated biometric trends (7-day avg. heart rate, sleep efficiency, activity calories), and rule-compliant baseline targets.

- **Targets (y):** The historical, successfully adhered-to calorie intake and macronutrient distribution from a cohort of similar users. "Successfully adhered-to" is defined as periods where the user consistently logged food and maintained or progressed toward their goal.

- **Hyperparameter Tuning:** Will be performed using Bayesian Optimization via the hyperopt library to maximize predictive accuracy and generalization.

7.3.7. Planner & Optimization

The Planner translates the final calorie and macronutrient targets from the ML layer into a practical daily meal plan.

- **Constraint-Based Optimization:** The planner solves an optimization problem that aims to:

    1. Meet the precise macronutrient and calorie targets.

    2. Distribute nutrients optimally across meals (e.g., higher protein breakfast for satiety, complex carbs
    3. around workout times).

    4. Adhere to all rules from the Multi-Condition Matrix (e.g., low sodium, low sugar).

    5. Incorporate user food preferences and allergies from their profile.

- **Output:** The output is a structured JSON object detailing each meal (breakfast, lunch, dinner, snacks), including suggested food items, portion sizes, and macronutrient breakdown per meal. This object is sent to the **Meal Generator** service to populate with specific recipes and ingredients.

7.3.8. Dynamic Recalculation & Trigger-Based Adjustments

Recalculation is not solely time-based (e.g., weekly); it is event-driven.

- **Significant Weight Change:** A weight update exceeding a pre-defined threshold (e.g., ±2% of body weight) triggers an immediate recalculation of TDEE and targets.

- **Biometric Shift Detection:** A **CUSUM (Cumulative Sum)** control chart monitors the time-series data of the user's weight and daily energy expenditure [32]. The CUSUM algorithm is effective at detecting small, persistent shifts in the mean of a process.

  When the CUSUM statistic exceeds a pre-defined threshold, it triggers a recalculation, indicating a significant change in the user's metabolic state (e.g., metabolic adaptation during weight loss, changed activity level).

- **User Feedback:** Explicit user feedback (e.g., marking a plan as "too hard" or "too easy") can also flag the system for a potential review.

7.3.9. Explainability & Audit

This module ensures the system's recommendations are transparent and trustworthy.

- **Technical Explainability (XAI):** The module uses **SHAP (SHapley Additive exPlanations)** values to determine the contribution of each feature (e.g., age, sleep, condition) to the final model output. This quantifies why the user's target is different from a baseline average.

- **Natural Language Generation (NLG):** A template-based system converts the technical insights from SHAP into user-friendly explanations.

    - Technical Insight: shap_values(age) = +150 kcal, shap_values(sleep_duration) = -100 kcal

    - Natural Language Output: "Your calorie target is slightly higher than average for your goal because of your younger age, which is partially offset by your relatively lower sleep duration from last week."

- **Audit Log:** A comprehensive log in the PostgreSQL database records every significant action: timestamp of calculation, inputs used, rules applied, model version, final outputs, and the generated explanation. This provides full traceability for debugging and regulatory compliance.

7.4. BIM-Inspired Digital Twin Integration

The system's performance will be evaluated through:

- **Offline Testing:** Using historical or synthetic data, evaluating model accuracy (MAE, RMSE) in predicting adhered-to macronutrient distributions.

- **A/B Testing:** Deploying the hybrid MDANE system against a baseline rule-only system for a cohort of users, measuring KPIs like adherence rate, goal achievement rate, and user satisfaction.

- **Clinical Validation:** Partnering with nutritionists to review generated plans for a subset of users with complex conditions, assessing clinical safety and appropriateness.

7.5. Technologies & Tech Stack

- **Backend:** FastAPI (for async capabilities and OpenAPI standards), Celery (for background tasks like recalculation), PostgreSQL with TimescaleDB, Redis (for caching and Celery broker).

- **ML Stack:** Python 3.9, Scikit-learn, XGBoost, Pandas, NumPy, SciPy, SHAP.

- **APIs:** Google Fit API, Apple HealthKit API, OAuth 2.0.

- **DevOps:** Docker, Docker Compose, CI/CD via GitHub Actions, deployment on AWS EC2.

- **Monitoring:** Prometheus and Grafana for monitoring API performance and model metrics.

8. Requirements

This chapter outlines the critical requirements for the development and successful deployment of the Multi-Disease Adaptive Nutrition Engine (MDANE). These requirements are categorized into functional (what the system must do), non-functional (how the system should perform), and user-centric needs.

8.1. Functional Requirements

ID	Requirement Description	Priority
FR1	The system shall allow users to create and manage a profile containing age, sex, height, weight, health goals, and medical conditions.	High
FR2	The system shall securely authenticate and connect with Google Fit and Apple HealthKit APIs to retrieve biometric data.	High
FR3	The system shall calculate Basal Metabolic Rate (BMR) using the Mifflin-St Jeor equation.	High
FR4	The system shall compute Total Daily Energy Expenditure (TDEE) using MET-based activity multipliers.	High
FR5	The system shall calibrate the estimated TDEE using a 7-day rolling average of Active Energy Expenditure (AEE) from wearable data.	High
FR6	The system shall apply constraints from the Multi-Condition Matrix to all nutritional recommendations.	High
FR7	The system shall employ an XGBoost model to personalize calorie and macronutrient targets based on user data and trends.	High
FR8	The system shall generate a daily meal plan (as nutrient targets) that adheres to all user constraints and personalized goals.	High
FR9	The system shall provide natural language explanations for its recommendations, citing influential factors.	Medium
FR4	The system shall compute Total Daily Energy Expenditure (TDEE) using MET-based activity multipliers.	High
FR5	The system shall calibrate the estimated TDEE using a 7-day rolling average of Active Energy Expenditure (AEE) from wearable data.	High
FR6	The system shall apply constraints from the Multi-Condition Matrix to all nutritional recommendations.	High

8.2. Non-Functional Requirements

ID	Category	Requirement Description
NFR1	Performance	The system must generate a meal plan within 3 seconds of a request 99% of the time.
NFR2	Security	All Personally Identifiable Information (PII) and health data must be encrypted at rest (AES-256) and in transit (TLS 1.3).
NFR3	Security	The system must be compliant with relevant data protection regulations (HIPAA, GDPR) for health information.
NFR4	Scalability	The system architecture must support scaling to 10,000 concurrent users without degradation of service.
NFR5	Reliability	The system must achieve 99.9% uptime for core services (API, calculation engine).
NFR6	Usability	The mobile application must achieve a System Usability Scale (SUS) score of >80 upon user testing.
NFR7	Maintainability	All code must be version-controlled (Git) and have a test coverage of at least 80% for core modules.
NFR8	Interoperability	The system's public API must adhere to OpenAPI (Swagger) 3.0 specifications for easy integration.

8.3. User requirements

1. As a user, I want to easily connect to my smartwatch so that my activity data is automatically synced and used.

2. As a user with diabetes, I need the meal plan to strictly limit added sugars and prioritize low-glycemic foods so that I can manage my blood glucose levels.

3. As a user, I want to understand *why* a specific calorie target or food suggestion was made so that I can trust the system and feel empowered.

4. As a user, I need to adapt automatically when my weight changes or my activity level significantly increases/decreases so that it remains effective.

5. As a user, I want a simple and intuitive interface to log my weight and see my progress over time so that I can stay motivated.

9. Commercialization & Potential Applications

The FitNourish AI system is designed to provide personalized, affordable, and practical nutrition guidance. Its initial core facility focuses on creating meal plans based on low-price items available at Food City, making healthy eating accessible and budget-friendly.

Initial Core Facility:

- ❖ Budget-Friendly Meal Planning: The system generates personalized meal plans using the most affordable and nutritious ingredients available at outlets. This ensures users receive a practical shopping list that aligns with their health goals and financial constraints.

Future Expansion Plans:

1. Multi-Store Integration:
   The system will expand to include other popular supermarkets (such as Keells, Arpico, and Lanka Sathosa) and local markets. This will provide users with more flexibility and choice, allowing them to compare prices and select the most convenient and cost-effective shopping options.

2. Integration with Medical Centers and Doctors:
   - ❖ Collaboration with Healthcare Providers: We plan to partner with medical centers, clinics, and doctors to integrate FitNourish AI into patient care plans. This would allow doctors to prescribe personalized nutrition plans directly to patients managing conditions like diabetes, hypertension, or obesity.
   - ❖ Remote Monitoring and Support: Healthcare providers could monitor patient adherence to nutritional guidelines and provide feedback through the platform, improving patient outcomes and engagement.
   - ❖ Continual Improvement: Feedback from medical professionals would

be used to refine the algorithm, ensuring recommendations remain aligned with the latest clinical guidelines and real-world needs.

3. Community and Social Features:
    - ❖ User Groups and Challenges: Introduce community-based features where users can join groups (e.g., weight loss, diabetes management) and participate in challenges to stay motivated.
    - ❖ Recipe Sharing: Allow users to share their own healthy, budget-friendly recipes, creating a crowdsourced database of affordable meal ideas.
4. Education and Workshops:
    - ❖ Nutritional Workshops: Collaborate with nutritionists and doctors to offer virtual workshops on topics like meal prep, understanding nutritional labels, and managing health conditions through diet.
    - ❖ Gamified Learning: Incorporate educational content into the app using quizzes, videos, and interactive tools to help users make informed choices.

9.1. Gantt Chart

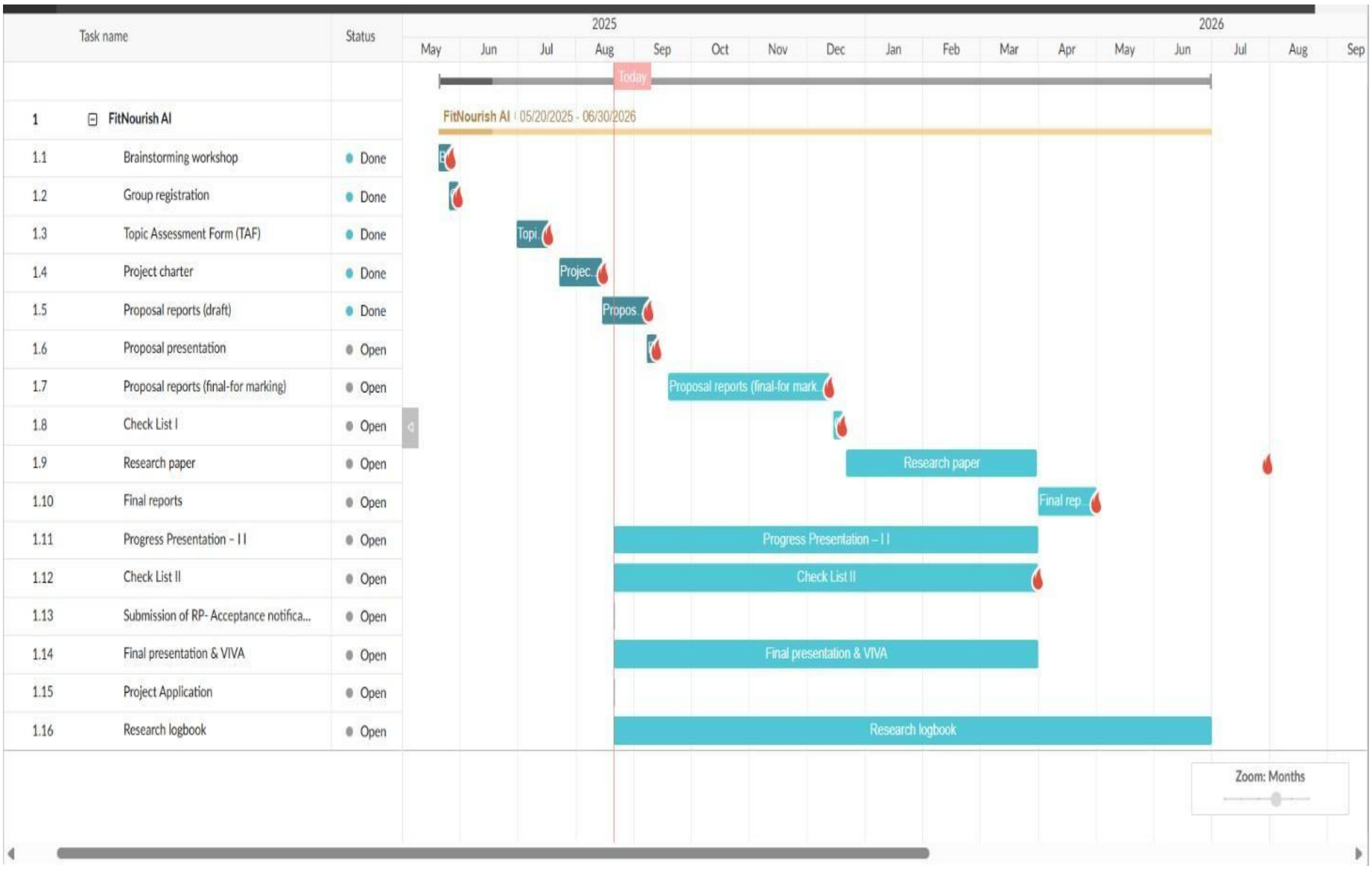

Figure 9. Gantt Chart

10. Description of Personal and Facilities

Personal:

- Principal Researcher: Responsible for overall project execution including literature review, system design, software development, data pipeline implementation, user study conduction, results analysis, and thesis writing. Possesses expertise in Python, machine learning (Scikit-learn, XGBoost), API development (FastAPI), and database management.

- Project Supervisor: Provides academic guidance, ensures research rigor, validates technical approach, facilitates resource access, and offers continuous feedback throughout project lifecycle.

- Clinical/Nutritional Advisor: Provides consultative expertise to ensure clinical safety and nutritional accuracy of the Multi-Condition Matrix, particularly for reviewing constraints derived from clinical guidelines.

Facilities:

- Computing Hardware: Access to high-performance computing laboratories and personal development workstations

- Software and Development Tools: Comprehensive suite of open-source technologies including Python, PostgreSQL, Docker, and machine learning libraries (Pandas, NumPy, Scikit-learn, XGBoost, SHAP)

- Cloud Services: Cloud computing credits for deployment and large-scale testing

- Research Resources: Access to digital research databases and academic publications

- Workspace: Dedicated meeting rooms and work areas for collaboration and focused work

11. Budget Estimation and Justification

A preliminary budget has been prepared to cover the essential resources required for the successful implementation and evaluation of the proposed system. Table 1 summarizes the anticipated costs.

Table 1: Estimated Budget

Category	Description	Estimated Cost (LKR)
Hardware / Devices	Samsung Galaxy Watch 6 – for biometric data acquisition (heart rate, SpO_2, sleep, stress monitoring).	50,000
Software / Tools	Python, TensorFlow, PyTorch, Google Colab (open-source / free tiers).	0
Cloud / Hosting	GPU-enabled cloud resources (Google Colab Pro / AWS EC2) for model training.	6,000
Datasets	Publicly available datasets (free) and premium access where necessary.	2,000
Documentation & Printing	Report preparation, printing, and binding.	3,000
Miscellaneous	Internet, electricity, and contingency costs.	4,000
Total Estimated Budget		65,000

Justification:

The primary cost arises from the acquisition of the Samsung Galaxy Watch 6, which will be used to collect real-time biometric data including heart rate, SpO_2 levels, stress, and sleep quality. This device was selected due to the high accuracy of its sensors, comprehensive health monitoring features, and SDK/API compatibility, which ensures seamless integration with the proposed nutrition and wellness system. Although lower-cost alternatives such as fitness bands exist, their limited sensor reliability and restricted developer support make them less suitable for research validation.

Other costs are minimal since most of the required software tools are open-source. However, limited expenditure is allocated for cloud-based GPU resources to accelerate deep learning model training, as well as for dataset acquisition, documentation, and contingency requirements.
Overall, the total budget of LKR 65,000 is justified, with the smartwatch representing the critical hardware investment necessary to ensure data accuracy, system feasibility, and reliable evaluation outcomes.

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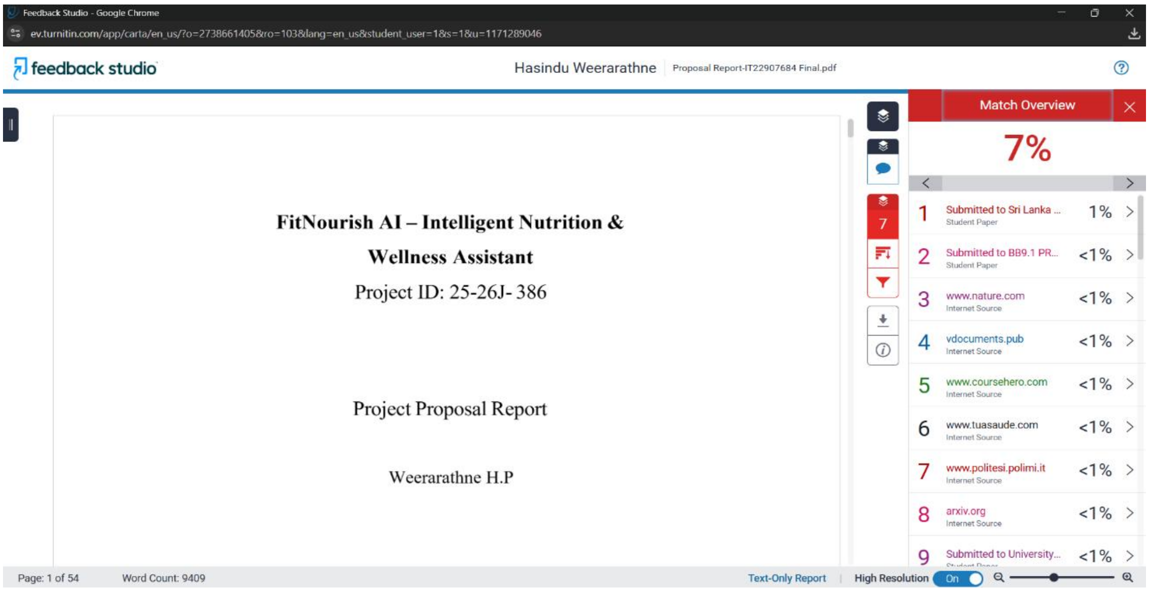
13.Appendix



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